



MOHAMMAD ROHAAN

BSCS Candidate · FAST NUCES, Islamabad

Software Engineering · AI/ML · Data Science · Robotics & Embedded Systems

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OBJECTIVE

Final-year BSCS student at FAST NUCES, Islamabad (June 2027). I build **full-stack systems** and **applied AI**: a live bilingual voice agent, NLP quiz generation, YOLOv8 detection, and a Spring Boot library platform, plus robotics/embedded prototypes on ESP32, Arduino, and Raspberry Pi. Seeking **software engineering, AI/ML, or embedded robotics internships** and **junior** roles.

EDUCATION

FAST NUCES — Islamabad, Pakistan

Bachelor of Science in Computer Science (BSCS) Expected June 2027

TECHNICAL SKILLS

Languages: Python, Java, C, C++, C#, JavaScript, SQL, PHP, Assembly x86

AI / ML / Data: Scikit-learn, pandas, NLP (TF-IDF), CNN / DL, YOLOv8, Streamlit, model evaluation, statistics & distribution analysis, data preprocessing, feature engineering

Software Eng: Java 17, Spring Boot 3.5, Spring MVC, Thymeleaf, Spring Security, Spring Data JPA (Java Persistence API), Hibernate, MySQL 8, Next.js, Scrum / Agile, REST APIs, Git, Maven

Web & Databases: HTML, CSS, JS, PHP, SQL Server, T-SQL, ER / EER modeling, normalization

Systems: OpenMP, parallel & distributed computing, multithreading, mutexes / semaphores, benchmarking, DSA, numerical computing (SpMV / HPC)

Embedded / Robotics: Arduino, ESP32, Arduino IDE, Raspberry Pi, ultrasonic & common sensors, MATLAB (robotics simulation & kinematics basics), control themes (course/projects)

Tools & Other: Linux, shell scripting, Cisco Packet Tracer

RELEVANT COURSEWORK

Data Structures & Algorithms, Operating Systems, Computer Networks, Database Systems, Software Engineering, Probability & Statistics, Parallel & Distributed Computing, Numerical Computing, Design & Analysis of Algorithms, Machine Learning / Data Science, Computer Vision.

SELECTED PROJECTS

AI Voice Cold-Calling Agent (Next.js, VAPI, Groq, Deepgram) —

Live EN/Urdu sales voice agent with recordings, transcripts, and call history; deployed on Vercel.

AI Quiz Generator (Python, Streamlit, scikit-learn) —

RACE reading-comprehension quizzes with SVM ranking, distractors/hints, and METEOR 0.69.

LibraryMS (Java 17, Spring Boot, MySQL, Scrum) —

Role-based library (Admin/Librarian/Student) with reservations, fines, and 22/22 test cases.

Shuttlecock Detection (YOLOv8, transfer learning) —

Fine-tuned on 15,000 images for robotic pickup; realtime webcam inference (Pi + Arduino design).

Hospital DB System (SQL Server, PHP, ER/EER) —

Normalized hospital schema, 12 operational queries, and PHP/JS staff portals.

K-means Triangle Inequality (C++, OpenMP) —

Elkan vs Lloyd experiments; up to 2.2x faster than naive K-means at 128-D.